J Shoulder Elbow Surg (2018) 27, 17–22

Journal of Shoulder and Elbow Surgery

www.elsevier.com/locate/ymse

Return to play after shoulder instability in National Football League athletes

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Hypothesis: We hypothesized that National Football League (NFL) players sustaining a shoulder destabilizing injury could return to play (RTP) successfully at a high rate regardless of treatment type.

Methods: We identified and evaluated 83 NFL players who sustained an in-season shoulder instability event while playing in the NFL. NFL RTP, incidence of surgery, time to RTP, recurrent instability events, seasons/games played after the injury, and demographic data were collected. Overall RTP was determined, and players who did and did not undergo operative repair were compared.

Results: Ninety-two percent of NFL players returned to NFL regular season play at a median of 0.0 weeks in those sustaining a shoulder subluxation and 3.0 weeks in those sustaining a dislocation who did not undergo surgical repair ($P = .029$). Players who underwent operative repair returned to play at a median of 39.3 weeks. Forty-seven percent of players had a recurrent instability event. For players who were able to RTP, those who underwent surgical repair (31%) had a lower recurrence rate (26% vs. 55%, $P = .021$) and longer interval between a recurrent instability event after RTP (14.7 vs. 2.5 weeks, $P = .050$).

Conclusion: There is a high rate of RTP after shoulder instability events in NFL players. Players who sustain shoulder subluxations RTP faster but are more likely to experience recurrent instability than those with shoulder dislocations. Surgical stabilization of the shoulder after an instability event decreases the chances of a second instability event and affords a player a greater interval between the initial injury and a recurrent event.

Level of evidence: Level III; Retrospective Cohort Design; Treatment Study

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Keywords: Shoulder dislocation; shoulder subluxation; NFL; return to play; professional athletes; shoulder injury

Glenohumeral joint instability is a common orthopedic condition for young athletes participating in contact sports, affecting thousands of collegiate athletes each year.[16] The unique anatomy of the shoulder provides inherent laxity that may present as a spectrum of disorders that range from minor subluxations to frank dislocations. For the competitive athlete, treatment options have significant implications for player performance and longevity.

Current treatments for shoulder instability include nonoperative therapies emphasizing a combination of immobilization and physical therapy[18] or operative procedures aimed at restoring the compromised stabilizing elements by direct repair or reconstruction.[19] Proposed benefits to surgical repair over conservative management include reduced rates

Institutional Review Board approval was not required for this retrospective cohort study that used publicly available Internet-based reports.

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https://doi.org/10.1016/j.jse.2017.07.027

of recurrent instability[1,8,11,17] and possible prevention of further joint damage.

Previous studies have demonstrated variability with respect to return to play (RTP) and athletic performance with each treatment type. For example, Dickens et al[5] demonstrated that 73% of collegiate athletes could successfully RTP after nonoperative management of a shoulder dislocation without any limitations. In contrast, Arciero et al[1] prospectively compared a nonoperative treatment regimen to operative repair in young military cadets and found significantly fewer recurrences in the operative group.

Although shoulder instability has been examined in athletes of varying skill sets in prior studies,[3,5,12,18] limited data are available regarding shoulder instability in elite American football athletes such as those in the National Football League (NFL).[3,9] The purpose of this study was to determine the RTP rate of NFL athletes who experience a shoulder instability event according to the type of instability and mode of treatment. A secondary aim was to identify factors at the time of the initial injury that predict a player's career longevity, ability to RTP, or incidence of recurrent instability. We hypothesized that NFL players sustaining a shoulder destabilizing injury could RTP successfully at a high rate regardless of treatment type and that those undergoing nonoperative treatment would be at greater risk for developing recurrent shoulder instability after successful RTP.

Materials and methods

We conducted a retrospective cohort study evaluating NFL players with a history of a shoulder instability event while playing professionally between 2006 and 2014. These players were identified by methods similar to those in previous studies that used publically available Internet-based reports.[6,7,10,13-15] Sources for injury reports included team injury reports, team Web sites, press releases, personal Web sites, and professional and college football statistical Web sites. Injuries were verified using www.NFL.com. A player was included in the instability cohort if he was drafted to the NFL and participated in at least 1 NFL game before sustaining a shoulder instability event.

For the purposes of this study, subluxation was defined as a transient instability requiring no true reduction maneuver or positive radiographic/magnetic resonance imaging. A dislocation was defined as complete loss of glenohumeral joint congruency and requiring a manual reduction maneuver or documented imaging of the dislocation. The direction of the instability could not be identified from the reports reviewed.

Demographic data for each player were recorded, including age, NFL experience, height, weight, and position. RTP data, including date of injury, overall RTP rate, and RTP on the same team were recorded. Also recorded were the number of regular season games, playoff games, and seasons played before and after injury, as well as career length, draft round, and number of Pro Bowl selections before and after injury. Return to same team was based on the team for which the athlete appeared in his first regular season game after injury. Athletes with a report of "undisclosed shoulder injury" or who were on the injured list at the time of data gathering were excluded. A thorough search was conducted of all included players to identify those who underwent surgical treatment during the season or in the off-season. The percentage of the season lost was calculated by dividing the number of regular season games lost because of injury by 16 (number of regular season games in a standard NFL season).

A control group was then identified to compare our cohort with a representative group of NFL players with similar attributes who had not sustained a shoulder instability event. The control group was assembled via a blinded age-, size-, and position-matched cohort of NFL players that had previously not sustained a shoulder instability injury. Players were also excluded if there was a report of "undisclosed shoulder injury." Our methods for selecting a control cohort were similar to those reported in previous literature.[6,7,10,13-15]

We then compiled a deidentified database of all remaining players who participated in the NFL between 2006 and 2014. Players were matched to the respective athletes by the year of the instability event for the respective player in the original cohort, designated as the index year. The controls were then selected based on age during the index year, qualified as ±1 year, listed position, NFL experience, height, and weight. Beyond demographic data, for each control player we evaluated career length, draft round, and number of Pro Bowl selections before and after injury in the index year.

Primary outcome measures included type of instability injury (subluxation/dislocation), initial treatment (surgery/no surgery), RTP, time to RTP (in weeks), presence of recurrent instability, time to recurrent instability from RTP date (in weeks), and treatment for recurrent instability, if applicable.

The χ^2 test or Fisher exact test were used for categoric variables and independent *t* test or Wilcoxon rank sum test for continuous data. The Kruskal-Wallis test was used to test for differences between categoric variables with greater than 2 groups and a continuous variable. Pearson correlations were used to test for differences between 2 continuous variables. A paired *t* test was used to test for differences before and after injury among NFL players at different time points. All analyses were done using SAS 9.4 software (SAS Institute Inc., Cary, NC, USA). Statistical significance was set at $P < .05$.

Results

We identified 83 NFL players who experienced a shoulder instability event from 2006 to 2014, with a dislocation injury occurring in 72 and subluxation in 11. Demographic characteristics of the study cohorts and control players are listed in Table I. Players were more likely to undergo surgical repair if they injured their right arm ($P = .027$) or were injured in the later part of the season ($P = .013$). Twenty-six players (31%) were treated operatively after their index instability episode, and the remaining 57 players (69%) were treated nonoperatively. Seventy-seven NFL players (92.8%) successfully RTP after an in-season shoulder instability event occurred, regardless of the initial treatment method. Of the players who RTP, 46% (36 of 77) later experienced a recurrent shoulder instability event; however, 94% were once again able to RTP.

Time required to RTP is listed in Table II. Players treated nonoperatively who sustained a shoulder subluxation RTP faster than those sustaining a shoulder dislocation (median, 0.0 vs. 3.0 weeks; $P = .029$). Irrespective of a player's instability type, RTP was significantly quicker for players who

Table I Demographic data

Variable*	Surgery	No surgery	*P* value†
Age, y	25.6 ± 2.3	26.1 ± 3.3	.482
Body mass index, kg/m^2	29.7 ± 4.2	28.8 ± 3.0	.900
Position			
Quarterback	5 (19.2)	5 (8.8)	
Wide receiver	4 (15.3)	5 (8.8)	
Guard	1 (3.8)	2 (3.5)	
Safety	4 (15.3)	13 (22.8)	
Linebacker	4 (15.3)	7 (12.3)	
Offensive tackle	2 (7.7)	0	
Cornerback	4 (15.3)	12 (21.1)	
Tight end	0	3 (5.3)	
Running back	1 (3.8)	8 (14.0)	
Defensive end	0	2 (3.5)	
Defensive tackle	1 (3.8)	0	
Date of injury‡			**.013**
Spring/preseason/week 4	10 (38.5)	28 (49.1)	
Week 5 to end of season	16 (61.5)	29 (50.9)	
Laterality			**.027**
Left	11 (44)	31 (59.6)	
Right	14 (56)	21 (40.4)	
Draft round (mean)			.318
≤Third round	18 (69.2)	35 (62.5)	
>Third round	8 (30.8)	21 (37.5)	

* Data are presented as the mean ± standard deviation or number (%).
† Bold values are statistically significant.
‡ Date of injury: Spring/preseason/week 4 vs. week 5 to end of season.

Table II Return to play

Variable	Players (No.)	Return to play (median weeks)	*P* value*
Date of injury†			**.004**
Spring/preseason/week 4	27	3.1	
Week 5 to end of season	28	0.5	
Side			.625
Left	29	2.0	
Right	21	3.0	
Injury type (nonoperative treatment)			**.029**
Dislocation	47	3.0	
Subluxation	10	0.0	
Treatment type			**<.001**
Operative	26	39.3	
Nonoperative	57	2.3	

* Bold values are statistically significant ($P < .05$).
† Date of injury: Spring/preseason/week 4 vs. week 5–end of season.

were injured in the late season than for those who were injured in the early season (median, 0.5 vs. 3.1 weeks; $P = .004$). Operative management required significantly longer recovery time than nonoperative treatment (median, 39.3 vs. 2.3 weeks; $P < .001$).

Factors effecting recurrent instability are presented in Figs. 1 and 2. There was no difference in the incidence of recurrent instability events between players sustaining a dislocation vs. subluxation. However, players in both groups who underwent surgical treatment were less likely to have a recurrent instability event (26.1%) than those treated nonoperatively (54.6%; $P = .022$). Of the players with recurrent instability episodes, those with shoulder subluxations experienced a recurrent instability event sooner than patients with shoulder dislocations (median, 1.5 vs. 5.0 weeks; $P = .044$). Patients who were treated operatively enjoyed a significantly longer recurrence-free interval than the nonoperative group (median, 14.7 vs. 2.5 weeks; $P = .050$).

Patients who were able to RTP exhibited a significant decrease in the average number of games played per season. RTP players played an average of 1.5 fewer games per season than in their preinjury state ($P = .014$). However, when comparing our injured cohort to matched NFL controls, we found no differences in games, seasons, or Pro Bowl selections after RTP (Table III).

Table III Playing time of athletes who return to play compared with matched controls

Variable*	RTP group	Controls	*P* value
Games			
Preinjury	44.7 ± 31.4	46.6 ± 30.4	.642
Post-RTP	40.9 ± 35.5	42.2 ± 32.4	.402
Seasons			
Preinjury	3.3 ± 2.2	3.5 ± 2.1	.368
Post-RTP	3.4 ± 2.3	3.3 ± 2.2	.891
Games played per season			
Preinjury	12.2 ± 3.6	12.4 ± 3.7	.543
Post-RTP	10.9 ± 3.8	11.6 ± 4.4	.182
Career	7.4 ± 3.5	7.5 ± 2.8	.506
Pro Bowl			
Preinjury	0.7 ± 1.8	0.3 ± 1.0	.100
Post-RTP	0.5 ± 1.6	0.4 ± 0.9	.842

RTP, return to play.
* Data are provided as the mean ± standard deviation.

Discussion

Although glenohumeral instability commonly occurs in the NFL, there is no single consensus regarding RTP criteria or initial treatment modalities. The factors effecting RTP and treatment implications have not been clearly defined. Our study found that 92.8% of players who experience a shoulder instability event in the NFL will RTP. For players treated nonoperatively, those having subluxations and who were injured in the later part of the season were likely to RTP faster than those who were not. Players treated surgically were less likely to sustain a recurrent instability event and played for a longer period before reinjury, if reinjury occurred. Players who RTP showed no difference compared with matched controls for overall career longevity; however, they did display

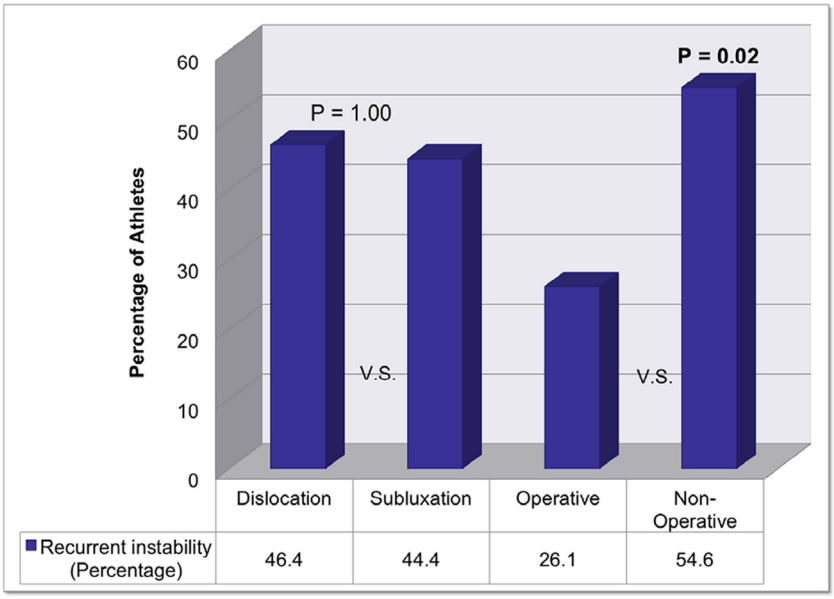

Figure 1 Incidence of recurrent instability. Percentage of athletes who sustained recurrent instability events compared by injury type and by treatment type. National Football League players treated nonoperatively experienced a higher incidence of recurrent instability than the surgical cohort. There was no difference in recurrent instability events between the dislocation and subluxation cohorts. Bold values are statistically significant.

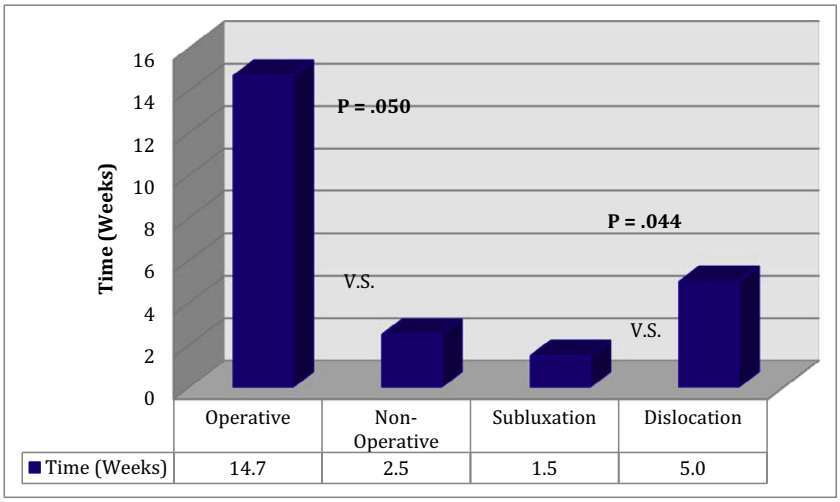

Figure 2 Time to recurrent instability event. Time after return to play from a primary instability event to recurrent instability event compared by treatment type and injury type. Data are reported by median weeks. National Football League players treated operatively had a significantly longer time to recurrent instability. Players with dislocation also had a significantly longer time to recurrent instability compared to subluxation. Bold values are statistically significant.

a significant decrease in the number of games played per season when returning to competition.

Several studies have evaluated RTP after shoulder instability at the high school or collegiate level. Dickens et al[5] demonstrated that 73% of athletes within the United States Military Academies RTP with nonoperative management after an in-season shoulder instability event. Buss et al[4] revealed an in-season RTP success rate of 90% in 30 high school and collegiate athletes. However, these studies were limited to high school and collegiate athletes, and limited information is available on NFL-level athletes. Our cohort of NFL athletes demonstrated a 93% RTP after a shoulder instability event, irrespective of treatment type. These results are similar to those reported for the non-professional athlete and suggest that shoulder instability events do not preclude athletes from continuing their playing career, regardless of skill level. The higher RTP rate in NFL players suggests a combination of factors, including player motivation, access to medical resources, and rehabilitation services.

Although nonoperative management leads to successful recoveries in some injuries, surgical management has been shown to decrease recurrence rates in the literature. LeClere et al[12] evaluated the medical records of a single NFL team and found that nonoperative treatment of shoulder instability led to significantly higher rates of repeat dislocation (41.7% vs. 10.5%; $P = .04$) and earlier time to recurrence (4.4 vs. 26 months). Zaremski et al[19] found in a large meta-analysis of young and adolescent athletes that nonoperative treatment had a greater than 10-fold chance of developing recurrent shoulder instability (odds ratio, 13.41) than if an operative stabilization was first performed.

Our study found that 46% of our RTP cohort experienced a secondary instability. However, upon stratifying for treatment type, patients who underwent surgical repair had a significantly reduced rate of recurrent shoulder instability than those who did not undergo surgical repair. Players who had surgical repair also played for a longer period before sustaining a recurrent instability event, if one occurred. These results demonstrate the protective benefit of undergoing surgical repair on recurrence rates and time to reinjury for the professional athlete.

For nonoperative patients, subluxation had a minimal effect, with players missing no playing time, whereas shoulder dislocation required a median of 3 weeks of recovery before RTP. Although players with shoulder subluxations are able to come back faster, there appears to be unintended drawbacks to their earlier return; that is, players with subluxations who RTP quicker appear to experience recurrent instability sooner. Likely an inadvertent consequence of an expedited RTP, the exact cause of this difference is unknown. One possible mechanism could involve inadequate periods of immobilization and periscapular physical therapy, which might limit the degree of total healing. These data show that balancing the short-term and long-term repercussions of shoulder instability are vital to appropriate counseling of patients after the development of an unstable shoulder.

Although the rate of successful RTP may be similar between operative and nonoperative approaches, duration until the RTP date is markedly different, an expected outcome given the requisite time needed for adequate recovery and postoperative rehabilitation after surgical stabilization. What is more intriguing about our results were the subtle effects a history of shoulder instability had on NFL career trends. In particular, injured players experienced a significant decrease in the average numbers of games played per season when their preinjury and postinjury states were compared. However there were no differences in athlete career longevity compared with the control population. This is contrasted by prior data collected from a single NFL team that demonstrated a significantly shorter career span for players with histories of shoulder instability[3] or those requiring shoulder stabilization surgery.[2] Additional epidemiologic studies on expected career length are needed and could further help clarify the unknown economic effect of the injury.

The effect of season timing of injury on RTP has not been previously evaluated. We found that individuals who sustained a shoulder destabilizing injury early within the season were slower to RTP than if the injury occurred later in the season. The basis for this phenomenon is unclear; however, one could speculate that the rate of RTP after an instability injury might be influenced by team needs at a particular time. That is, a player is more likely to be held out from playing activities early in the season to allow a more complete recovery vs. quicker return in order to be available for the last and critical games of the season (games to qualify for playoffs or playoffs themselves). Regardless, additional studies are needed to clarify how injury management is influenced by season timing.

Several limitations should be noted regarding this study. First, the data cohort used for the analysis were collected and pooled from an independent third-party source (NFL website), which lends itself to possible inaccuracies. This also allows for variations within the reported data resulting from individual orthopedic providers' management of each injury.

Another weakness of this study is the limited application it provides when translating the results to the general population. Although this was implicit to the study objective, the overall consistency of our data with previous studies on amateur athletes demonstrates that the general principles from our study data remain largely applicable. In addition, our cohort most likely does not include every player that experienced a shoulder instability episode during the study interval, which could lead to a selection bias. Also, prior medical history of shoulder instability is not always reported.

Finally, the direction of instability and the type of surgical repair were not collected. Direction of instability and the surgical procedure performed can influence the time to RTP and likelihood of a recurrent episode.

Conclusion

There is a high rate of RTP after shoulder instability events in NFL players. Players sustaining subluxations RTP faster but are more likely to have recurrent instability events than those who have dislocations. Surgical stabilization of the shoulder after an instability event decreases the chances of a second instability event and affords a player a greater interval between a recurrent event. Players sustaining a shoulder instability event may play in fewer games per season compared with before injury.

Disclaimer

The authors, their immediate families, and any research foundations with which they are affiliated have not received any financial payments or other benefits from any commercial entity related to the subject of this article.

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